Exercise 1.

Let

$$f(x) = \begin{cases} \frac{1}{3}e^{-x}, & \text{if } x < 0 \\ \frac{1}{3}, & \text{if } 0 \leq x < 1 \\ \frac{1}{3}e^{x}, & \text{if } 1 \leq x \end{cases}$$

a. Show that f is a valid probability density function.

b. Find $E(X)$

Exercise 2.

For some constant c, the random variable X has the probability density function

$$f(x) = \begin{cases} cx^4, & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

Find (a) $E(X)$ (b) $Var(x)$.

Exercise 3.

The random variable X has the probability density function

$$f(x) = \begin{cases} ax + bx^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

If $E(X) = 0.6$, find (a) $P\left(X < \frac{1}{2}\right)$ (b) $Var(X)$

Exercise 4.

Suppose that the cumulative distribution function of the random variable X is given by

$$F(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-x^2}, & x > 0 \end{cases}$$

Evaluate (a) $P(X > 2)$ (b)$P(1 < X < 3)$ (c) $E(X)$ (d) $Var(X)$

Exercise 5.

Let,

$$f(x) = \begin{cases} 12x^2(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find (a) Find the cdf of X (b)Find $P\left(0 < X < \frac{1}{2}\right)$ (c) $E(X)$ and $Var(X)$